

LAB 15

Scenario

The IT Security manager has now tasked you, the only Threat hunter, with performing multiple hunts regarding WMI attacks, Parent Process spoofing and Access Token theft.

The hunts will be conducted against Windows, PowerShell, and Sysmon Events of an endpoint that was compromised during a Red Teaming engagement. The abovementioned attacks all took place on that endpoint.

Once you successfully conclude your hunts, you can inform the rest of the blue team about the "traces" these attacks leave behind so that they can also hunt for them.

Goals

The learning objective of this lab is increase familiarity of detecting common techniques utilized by attacks by doing a hands-on analysis.

What you will learn

You will learn how to utilize available logs to detect certain threats on a compromised host.

Recommended tools & Resources

- PowerShell
- Python
- <https://attack.mitre.org/techniques/T1047/>
- <https://attack.mitre.org/techniques/T1502/>
- <https://attack.mitre.org/techniques/T1134/>

Tasks

Task 1. Have malicious WMI commands been executed on the machine?

You were informed that the red team utilized [WMImplant](#) during the red teaming engagement.

Since this implant utilizes WMI, you should begin by hunting for WMI spawning suspicious processes (such as PowerShell). Then, utilize publicly available reports on WMImplant to identify more artifacts/traces this implant leaves behind. Such a report is the below. https://cyberwardog.blogspot.com/2017/03/chronicles-of-threat-hunter-hunting-for_26.html

Be reminded that this hunt will be based on the available Windows, PowerShell and Sysmon events of the compromised endpoint.

Once you are done with this hunt, you can inform the blue team so that they can proactively hunt for such malware from there on.

Task 2. Has Parent Process Spoofing occurred on the machine?

You were informed that the red team utilized the mspaint.exe process while attacking this endpoint. You were also informed that they tried to conceal their activities by employing an evasion technique known as Parent Process Spoofing.

You are tasked with hunting for Parent Process Spoofing against the mspaint.exe process.

Parent Process Spoofing will be analyzed thoroughly in another lab. For the time being focus on the below.

- 6/28/2022 11:53:08 AM - Sysmon EventID 1 related to mspaint.exe
- 6/28/2022 11:53:08 AM - Sysmon EventID 8 related to mspaint.exe

SOLUTIONS

Below, you can find solutions for each task. Remember though, that you can follow your own strategy, which may be different from the one explained in the following lab.

Task 1. Have malicious WMI commands been executed on the machine?

After reading technical reports (a.k.a tactical threat intelligence) on WMImplant, we can identify the below artifacts/traces.

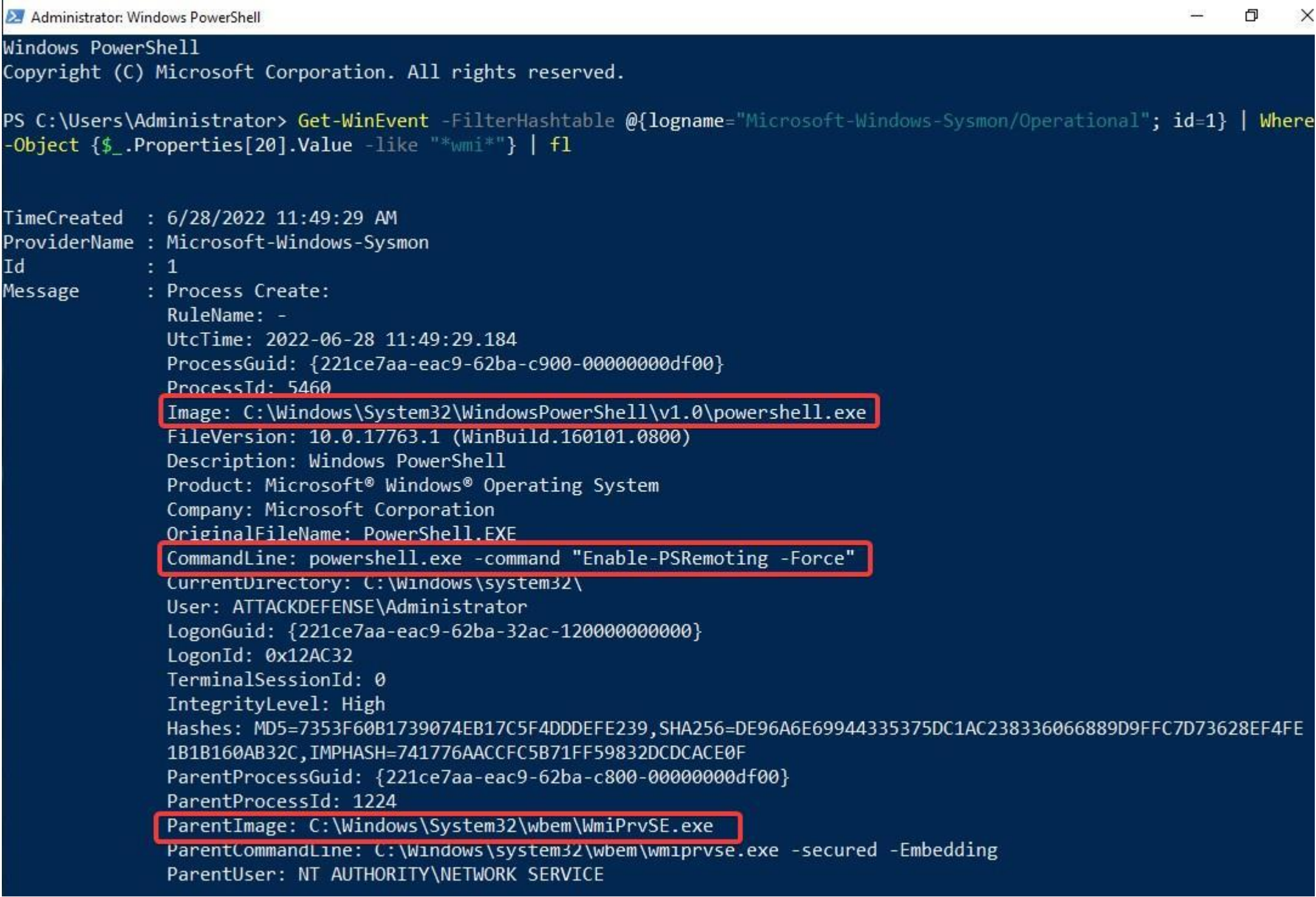
- wmiprsve.exe spawning powershell.exe and probably executing obfuscated commands

- Environment Variable name on EIDs 12 and 13; Pattern: [\\Environment\\\[a-zA-Z0-9\]{5}](#)
- by PowerShell.exe

Now for our hunt, we can take an approach where we either group the indicators above, or we do a search based only on one of them. Since we are dealing with a single source of logs, we can take the second approach (this can be tedious though when dealing with big environments and multi terabytes of logs). Let's begin by the first indicator and check if any processes have been spawned by "wmiprvse.exe". To parse for that, run the following command (inside a PowerShell as an administrator terminal):

```
Get-WinEvent -FilterHashtable @{logname="Microsoft-Windows-Sysmon/Operational"; id=1} | Where-Object {$_.Properties[20].Value -like "*wmi*"} | fl
```

The output of the command contains only 1 result, which looks as follows:



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> Get-WinEvent -FilterHashtable @{logname="Microsoft-Windows-Sysmon/Operational"; id=1} | Where-Object {$_.Properties[20].Value -like "*wmi*"} | fl

TimeCreated      : 6/28/2022 11:49:29 AM
ProviderName     : Microsoft-Windows-Sysmon
Id               : 1
Message          : Process Create:
                   RuleName: -
                   UtcTime: 2022-06-28 11:49:29.184
                   ProcessGuid: {221ce7aa-eac9-62ba-c900-00000000df00}
                   ProcessId: 5460
                   Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
                   FileVersion: 10.0.17763.1 (WinBuild.160101.0800)
                   Description: Windows PowerShell
                   Product: Microsoft® Windows® Operating System
                   Company: Microsoft Corporation
                   OriginalFileName: PowerShell.EXE
                   CommandLine: powershell.exe -command "Enable-PSRemoting -Force"
                   CurrentDirectory: C:\Windows\system32\
                   User: ATTACKDEFENSE\Administrator
                   LogonGuid: {221ce7aa-eac9-62ba-32ac-120000000000}
                   LogonId: 0x12AC32
                   TerminalSessionId: 0
                   IntegrityLevel: High
                   Hashes: MD5=7353F60B1739074EB17C5F4DDDEF239,SHA256=DE96A6E69944335375DC1AC238336066889D9FFC7D73628EF4FE
                           1B1B160AB32C,IMPHASH=741776AACCFC5B71FF59832DCDCACE0F
                   ParentProcessGuid: {221ce7aa-eac9-62ba-c800-00000000df00}
                   ParentProcessId: 1224
                   ParentImage: C:\Windows\System32\wbem\WmiPrvSE.exe
                   ParentCommandLine: C:\Windows\system32\wbem\wmiprvse.exe -secured -Embedding
                   ParentUser: NT AUTHORITY\NETWORK SERVICE
```

In fact, it appears that WmiPrvSE.exe has been executed on the machine, which spawns PowerShell with a command line parameter that aims to enable PowerShell Remoting.

We can "chain" that with Windows Security Logs, where we can see a login of the "Administrator" account from a remote machine with IP "13.229.91.59":

The screenshot displays the Windows Security Event Viewer interface. At the top, a header bar indicates 'Security' with 'Number of events: 335 (!) New events available'. Below this is a table of security events. The fifth row is highlighted in blue and has its timestamp, '6/28/2022 11:47:39 AM', circled in red. This row corresponds to Event ID 4624, categorized as 'Logon'. Below the table, the 'Event 4624, Microsoft Windows security auditing.' details pane is open. It has two tabs: 'General' and 'Details'. The 'General' tab is active, showing 'Network Information' with a red box around the 'Workstation Name: ATTACKER', 'Source Network Address: 13.229.91.59', and 'Source Port: 1289'. Below this, 'Detailed Authentication Information' shows 'Logon Process: NtLmSsp', 'Authentication Package: NTLM', 'Transited Services: -', 'Package Name (NTLM only): NTLM V2', and 'Key Length: 128'. A descriptive paragraph follows: 'This event is generated when a logon session is created. It is generated on the computer that was accessed. The subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe. The logon type field indicates the kind of logon that occurred. The most common types are 2 (interactive) and 3 (network). The New Logon fields indicate the account for whom the new logon was created, i.e. the account that was logged on.' At the bottom, a summary section lists: 'Log Name: Security', 'Source: Microsoft Windows security', 'Event ID: 4624', 'Level: Information', 'User: N/A', 'OpCode: Info', 'Logged: 6/28/2022 11:47:39 AM', 'Task Category: Logon', 'Keywords: Audit Success', 'Computer: AttackDefense', and a link for 'More Information: Event Log Online Help'.

Keywords	Date and Time	Source	Event ID	Task Category
Audit Success	6/28/2022 11:47:40 AM	Microsoft Windows security auditing.	4776	Credential Validation
Audit Success	6/28/2022 11:47:40 AM	Microsoft Windows security auditing.	4624	Logon
Audit Success	6/28/2022 11:47:40 AM	Microsoft Windows security auditing.	4672	Special Logon
Audit Success	6/28/2022 11:47:40 AM	Microsoft Windows security auditing.	4776	Credential Validation
Audit Success	6/28/2022 11:47:39 AM	Microsoft Windows security auditing.	4624	Logon
Audit Success	6/28/2022 11:47:39 AM	Microsoft Windows security auditing.	4672	Special Logon
Audit Success	6/28/2022 11:47:39 AM	Microsoft Windows security auditing.	4776	Credential Validation
Audit Success	6/28/2022 11:46:08 AM	Eventlog	1102	Log clear

Event 4624, Microsoft Windows security auditing.

General Details

Network Information:

Workstation Name: ATTACKER
Source Network Address: 13.229.91.59
Source Port: 1289

Detailed Authentication Information:

Logon Process: NtLmSsp
Authentication Package: NTLM
Transited Services: -
Package Name (NTLM only): NTLM V2
Key Length: 128

This event is generated when a logon session is created. It is generated on the computer that was accessed.

The subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.

The logon type field indicates the kind of logon that occurred. The most common types are 2 (interactive) and 3 (network).

The New Logon fields indicate the account for whom the new logon was created, i.e. the account that was logged on.

Log Name: Security
Source: Microsoft Windows security
Event ID: 4624
Level: Information
User: N/A
OpCode: Info
Logged: 6/28/2022 11:47:39 AM
Task Category: Logon
Keywords: Audit Success
Computer: AttackDefense
More Information: [Event Log Online Help](#)

After which, the PowerShell Process is spawned:

Security Number of events: 335 (!) New events available

Keywords	Date and Time	Source	Event ID	Task Category
Audit Success	6/28/2022 11:49:29 AM	Microsoft Windows security auditing.	4688	Process Creation
Audit Success	6/28/2022 11:49:29 AM	Microsoft Windows security auditing.	4688	Process Creation
Audit Success	6/28/2022 11:49:29 AM	Microsoft Windows security auditing.	4688	Process Creation
Audit Success	6/28/2022 11:48:59 AM	Microsoft Windows security auditing.	4719	Audit Policy Change
Audit Success	6/28/2022 11:48:59 AM	Microsoft Windows security auditing.	4719	Audit Policy Change
Audit Success	6/28/2022 11:48:59 AM	Microsoft Windows security auditing.	4719	Audit Policy Change
Audit Success	6/28/2022 11:48:59 AM	Microsoft Windows security auditing.	4719	Audit Policy Change
Audit Success	6/28/2022 11:48:59 AM	Microsoft Windows security auditing.	4719	Audit Policy Change

Event 4688, Microsoft Windows security auditing.

General Details

Security ID: NULL SID
Account Name: Administrator
Account Domain: ATTACKDEFENSE
Logon ID: 0x12AC32

Process Information:

New Process ID: 0x1554
New Process Name: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Token Elevation Type: %1936
Mandatory Label: Mandatory Label\High Mandatory Level
Creator Process ID: 0x4c8
Creator Process Name: C:\Windows\System32\wbem\WmiPrivSE.exe
Process Command Line:

Token Elevation Type indicates the type of token that was assigned to the new process in accordance with User Account Control policy.

Type 1 is a full token with no privileges removed or groups disabled. A full token is only used if User Account Control is disabled or if the user is the built-in Administrator account or a service account.

Type 2 is an elevated token with no privileges removed or groups disabled. An elevated token is used when User Account Control is enabled and the user chooses to start the program using Run as administrator. An elevated token is also used when an application is configured to always require administrative privilege or to always require maximum privilege, and the user is a member of the Administrators group.

Log Name: Security
Source: Microsoft Windows security Logged: 6/28/2022 11:49:29 AM
Event ID: 4688 Task Category: Process Creation
Level: Information Keywords: Audit Success
User: N/A Computer: AttackDefense
OpCode: Info
More Information: [Event Log Online Help](#)

Note - If you look for other artifacts that we listed earlier, you may not find any of them because the red team did not utilize all of WMIImplant's capabilities - perhaps all that was required was a single action enabling PSRemoting. **We would have missed this if we had searched only for occurrence of, for example, 2 out of the 3 detection rules.** Have this in mind during your hunts. Don't always search for multiple indicators at once!

Task 2. Has Parent Process Spoofing occurred on the machine?

By consulting with Sysmon's Process Creation logs or with the Security Process Creation logs, we can identify the following.

Operational Number of events: 659 (!) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	6/28/2022 11:53:08 AM	Sysmon	1	Process Create (rule: ...)
Information	6/30/2022 7:15:10 AM	Sysmon	1	Process Create (rule: ...)
Information	6/28/2022 11:53:05 AM	Sysmon	1	Process Create (rule: ...)

Event 1, Sysmon

General Details

Process Create:
 RuleName: -
 UtcTime: 2022-06-28 11:53:07.996
 ProcessGuid: {221ce7aa-eba3-62ba-0a01-00000000df00}
 ProcessId: 2324
 Image: C:\Windows\System32\mspaint.exe
 FileVersion: 10.0.17763.1 (winBuild.160101.0000)
 Description: Paint
 Product: Microsoft® Windows® Operating System
 Company: Microsoft Corporation
 OriginalFileName: MSPAINT.EXE
 CommandLine: 0
 CurrentDirectory: C:\Users\Administrator\Desktop\Tools\ppid-spoofing\
 User: ATTACKDEFENSE\Administrator
 LogonGuid: {221ce7aa-e948-62ba-ecde-020000000000}
 LogonId: 0x2DEEC
 TerminalSessionId: 2
 IntegrityLevel: High
 Hashes: MD5=67C68B11E98970966DF59D2FAD6152BF, SHA256=615CFFE98CAD0DB5F7F261CE915F13BBBC22378BB2A80591D38205D5658A8092, IMPHASH=ABBE6AE1A46B5D03FCCC5A2C2F1DF4D0
 ParentProcessGuid: {221ce7aa-e950-62ba-6a00-00000000df00}
 ParentProcessId: 4412
 ParentImage: C:\Windows\explorer.exe
 ParentCommandLine: C:\Windows\Explorer.EXE
 ParentUser: ATTACKDEFENSE\Administrator

Log Name: Microsoft-Windows-Sysmon/Operational
 Source: Sysmon Logged: 6/28/2022 11:53:08 AM
 Event ID: 1 Task Category: Process Create (rule: ProcessCreate)
 Level: Information Keywords:
 User: SYSTEM Computer: AttackDefense
 OpCode: Info
 More Information: [Event Log Online Help](#)

explorer.exe seems to be starting mspaint.exe. This looks quite natural, but an interesting event occurs immediately right after, where PowerShell creates a remote thread to mspaint.exe (which, to some extent, indicates that there is some kind of interaction between the 2 processes):

Operational Number of events: 670 (!) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	6/28/2022 11:55:31 AM	Sysmon	11	File created (rule: File
Information	6/28/2022 11:53:08 AM	Sysmon	8	CreateRemoteThread
Information	6/30/2022 7:15:08 AM	Sysmon	6	Driver loaded (rule: C
Information	6/30/2022 7:15:05 AM	Sysmon	6	Driver loaded (rule: C
Information	6/30/2022 7:15:09 AM	Sysmon	6	Driver loaded (rule: C
Information	6/30/2022 11:54:30 AM	Sysmon	5	Process terminated (rule: C

Event 8, Sysmon

General Details

CreateRemoteThread detected:
RuleName: -
UtcTime: 2022-06-28 11:53:08.726
SourceProcessGuid: {221ce7aa-eb75-62ba-0401-00000000df00}
SourceProcessId: 1932
SourceImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
TargetProcessGuid: {221ce7aa-eba3-62ba-0a01-00000000df00}
TargetProcessId: 3324
TargetImage: C:\Windows\System32\mspaint.exe
NewThreadId: 5548
StartAddress: 0x00007FFF7CD1F220
StartModule: -
StartFunction: -
SourceUser: ATTACKDEFENSE\Administrator
TargetUser: ATTACKDEFENSE\Administrator

KEY TAKEAWAYS

1. Robust detection rules and non-default audit policies are required to perform high quality hunts
2. Adversaries often choose to hide in plain sight
3. Defender tools can be abused or disabled by attackers